

Picture (Im-) Perfect

Exploring Imperfections in Computer Generated Rendering

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ABSTRACT

Computer graphics advances rapidly advance new innovations in animation practices aiming for ever more realism - calculating exact symmetry, perfect lines and flawless shadows that follow the physicality of reality. However, these technical advancements present a departure from traditional drawing techniques and physical artistry and handcraft, which often embrace originality, experimentation, and individual authorship in animation practices. This project explores hand-drawing and hand-painting techniques in the context of 3D shading and texturing, reintroducing a much-needed return to traditional artistic skills through a mixture of hand-painting and procedural texturing to 3D shading. Reaching beyond hyper-realism, this illustrates a new synthesis between 2D drawing and 3D shading techniques.

CCS CONCEPTS

• Computer graphics; • Rendering; • Non-photorealistic rendering;

KEYWORDS

Traditional Medium, Digital Medium Painting, Texturing

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1 INTRODUCTION

This project attempts to preserve lost qualities of artistic production predating the era of standardized contemporary 3D animation. To escape the confines of computed perfections, the 3D scene explores new modes of hand painted shading techniques. Generated procedural textures explore art styles which follow artistic examples but are emulated through unconventional, non-photorealistic rendering (NPR) approaches that mimic traditional mediums such as ink, oil paint, and pencil.

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Figure 1: Jolie Che (self). (2024) Ink Wash Study.

2 APPROACH

2.1 Overview

Achieving the desired effects in 3D space involves experimenting with NPR textures. To achieve a painterly result, a painterly reference is followed to guide the 3D shading process. In this instance, NPR techniques in Blender were explored: Alongside research and developing shaders for each medium (ink, oil paint, and pencil), a painted reference served as point of departure. Those references support direct comparison during rendering processes.

2.2 Reference Study and Painting

For Chinese ink painting studies, artists often use quick strokes of ink that capture the gestures of their subjects, along with washes of colored ink to add a pop of colour. After conducting a study of masters of this medium [Fig. 1], it was concluded that one of the most identifiable features of Chinese ink paint is in the gestural brush strokes.

For oil paint studies, directly referencing the famous *The Swing* [Fragonard, 1767–1768] [Fig. 2], the Rococo painter is observed to harbor complex composition and masterful rendering. While Post-Impressionist oil painters construct their paintings through rough and thick layers of pigment by using a palette knife to apply thick



Figure 2: (from left to right) [Fragonard. 1767–1768.] [Jolie Che (self). (2024) Oil Paint Study.]

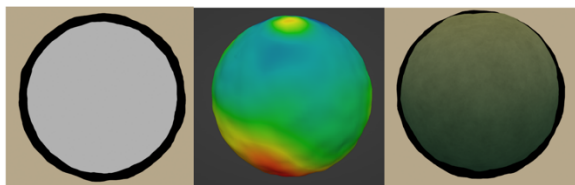


Figure 3: (Left to right) Outline, weight paint and final experiment

layers of pigment and dramatic tonal contrasts that form distinctive dimensions.

2.3 Ink Wash Experiment

The ink outlines were simulated by displacing a clone of the primary sphere. [Fig. 3] A noise texture was then applied and tweaked to the mesh-clone to simulate line weight variation. [Tada, 2021] Next, to simulate the tapering and gaps from weight pressure when artists' hands lift from the paper, weight painting was assigned to vertex groups of the clone. While weight painting is usually utilized in rigging to distribute weights to each assigned bone, here weight painting was applied to the outline clone's vertex groups, mimicking line weight. Lastly, a procedurally generated green gradient material was applied to the base mesh for aesthetic impact.

2.4 Oil Paint Experiment

The normal maps were baked and hand-painted over the normal map to apply to a highly reflective sphere [Fig. 4] [Gindy, 2023]. Several light sources were then introduced and adjusted around the mesh to optimize the light reflection on the mesh. A gradient map was then applied on top of the gradient map to produce an image texture that provides better visual representations for each normal. Lastly, the background was hand-painted for aesthetic impact.

2.5 Pencil Sketch Experiment

In this final case study, a hand-sketched pencil-hatching texture pattern was first procedurally applied to the sphere, the resulting material would only affect the dark area of the mesh [Fig. 5].



Figure 4: (Left to right) Normal Map painting, texture painting and background painting.

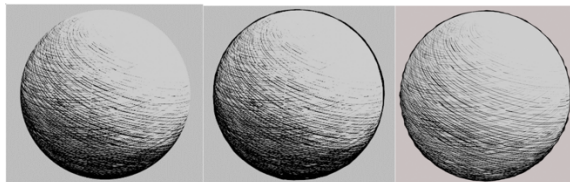


Figure 5: (Left to right) Patterned texture, clone outline and Grease Pencil outline

Then two layers of varied outlines were implemented to mimic the illusion of hand-sketched drawings. The first cloned mesh outline was a similar one from Ink Wash Experiment, but without weight-painting and adjusted to be thinner, mimicking the thin and short strokes of pencil drawings. A Grease Pencil layer was then added on top of the first layer of line, a feature in Blender that procedurally produces outlines of the selected mesh from the point of the camera, this is to mimic the messy multiple lines when sketching. [Littlerolz, 2022]

3 CONCLUSION

The experiments presented here shed new light onto NPR techniques, illustrating a new method to reintroduce hand-painted qualities to 3D rendering and shading. An unexpected amount of digital manipulation and iterations of layered effects was needed for these textures to match their traditional medium counterparts. This rendering method ironically proved more complex than simply painting with pen and paper. Results were achieved through several stages of iterative design, experimentation with coding nodes, as well as hand painting the textures.

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